KEALEE PLATFORM - SPRINT EXECUTION ROADMAP

Complete Build Instructions: Sprints 5-12 + Launch

Date: December 12, 2025

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CURRENT STATUS SUMMARY

✅ COMPLETED (Sprints 1-4):

- Backend: 60+ endpoints, 15+ models, production-ready
- Database: PostgreSQL running (localhost:5432/kealee_dev)
- Frontend: 70% complete (13 pages partially built)
- Authentication: Working
- Basic project management: Working

🎯 REMAINING WORK (Sprints 5-12):

- Sprint 5: Stripe Connect integration (escrow) - 7-10 days
- Sprint 6: Smart deposit verification - 5-7 days
- Sprint 7: Contractor vetting system - 3-4 days
- Sprint 8: AI milestone verification - 5-6 days
- Sprint 9: Frontend completion - 5-7 days
- Sprint 10: Testing & bug fixes - 3-5 days
- Sprint 11: Pre-launch prep - 2-3 days
- Sprint 12: LAUNCH! - 1 day

TOTAL REMAINING TIME: 31-43 days (4.5-6 weeks)

SPRINT 5: STRIPE CONNECT INTEGRATION (ESCROW)

Duration: 7-10 days | Priority: CRITICAL

OBJECTIVE: Escrow system working with third-party (Stripe)

DAY 1: STRIPE SETUP (2-3 hours)

☐ Create Stripe account: https://stripe.com
☐ Enable Stripe Connect in dashboard
☐ Get test API keys
  - Settings → API keys → Reveal test key
  - Copy publishable key (pk_test_...)
  - Copy secret key (sk_test_...)

☐ Install Stripe SDK:
```bash
cd /path/to/kealee-backend
npm install stripe @stripe/stripe-js
```

☐ Add to .env file:
```
STRIPE_SECRET_KEY=sk_test_your_key_here
STRIPE_PUBLISHABLE_KEY=pk_test_your_key_here
STRIPE_WEBHOOK_SECRET=(get this later)
```

☐ Create Stripe service file:
```javascript
// backend/src/services/stripe.service.js
const Stripe = require('stripe');
const stripe = Stripe(process.env.STRIPE_SECRET_KEY);

module.exports = { stripe };
```

☐ Test connection:
```javascript
const { stripe } = require('./services/stripe.service');

async function test() {
  const balance = await stripe.balance.retrieve();
  console.log('✅ Stripe connected!', balance);
}
test();
```

Expected: "✅ Stripe connected!"

DAY 2: CONTRACTOR STRIPE ACCOUNTS (4-5 hours)

☐ Add Stripe fields to contractors table:

```sql
ALTER TABLE contractors
ADD COLUMN stripe_account_id VARCHAR(255),
ADD COLUMN stripe_account_status VARCHAR(50),
ADD COLUMN stripe_onboarded_at TIMESTAMP;
```

☐ Create contractor onboarding endpoint:

```javascript
// POST /api/contractors/stripe-onboard
router.post('/stripe-onboard', async (req, res) => {
  const { contractorId, businessName, email, ein } = req.body;

  const account = await stripe.accounts.create({
    type: 'custom',
    country: 'US',
    email: email,
    capabilities: {
      card_payments: { requested: true },
      transfers: { requested: true }
    },
    business_type: 'company',
    company: {
      name: businessName,
      tax_id: ein
    },
    business_profile: {
      mcc: '1522' // Construction contractor
    }
  });

  await db.query(
    'UPDATE contractors SET stripe_account_id = $1, stripe_account_status = $2 WHERE id = $3',
    [account.id, 'pending', contractorId]
  );

  const accountLink = await stripe.accountLinks.create({
    account: account.id,
    refresh_url: `${process.env.APP_URL}/contractor/stripe/refresh`,
    return_url: `${process.env.APP_URL}/contractor/stripe/complete`,
```

```
    type: 'account_onboarding'
  });

  res.json({ onboardingUrl: accountLink.url });
});
```

☐ Frontend: Add "Connect Bank" button
☐ Test: Create test contractor, complete onboarding
☐ Verify stripe_account_id saved

DAY 3-4: ESCROW FUNDING (6-8 hours)

☐ Add escrow fields to projects table:
```sql
ALTER TABLE projects
ADD COLUMN escrow_payment_intent_id VARCHAR(255),
ADD COLUMN escrow_status VARCHAR(50),
ADD COLUMN escrow_amount DECIMAL(10,2),
ADD COLUMN escrow_balance DECIMAL(10,2),
ADD COLUMN escrow_funded_at TIMESTAMP;
```

☐ Create transactions table:
```sql
CREATE TABLE transactions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id UUID REFERENCES projects(id),
  milestone_id UUID,
  type VARCHAR(50),
  amount DECIMAL(10,2),
  status VARCHAR(50),
  payment_intent_id VARCHAR(255),
  transfer_id VARCHAR(255),
  metadata JSONB,
  created_at TIMESTAMP DEFAULT NOW()
);
```

☐ Create escrow funding endpoint:
```javascript
```

```
// POST /api/projects/:projectId/fund-escrow
router.post('/:projectId/fund-escrow', async (req, res) => {
  const { projectId } = req.params;
  const { paymentMethodId } = req.body;

  const project = await getProject(projectId);
  const contractor = await getContractor(project.contractor_id);

  const paymentIntent = await stripe.paymentIntents.create({
    amount: project.total_amount * 100,
    currency: 'usd',
    capture_method: 'manual', // Don't capture - hold in escrow
    metadata: {
      projectId: projectId,
      purpose: 'escrow'
    },
    application_fee_amount: Math.round(project.total_amount * 0.025 * 100),
    on_behalf_of: contractor.stripe_account_id,
    payment_method: paymentMethodId,
    confirm: true
  });

  await db.query(
    `UPDATE projects SET
     escrow_payment_intent_id = $1,
     escrow_status = $2,
     escrow_amount = $3,
     escrow_balance = $4,
     escrow_funded_at = NOW()
     WHERE id = $5`,
    [paymentIntent.id, 'FUNDED', project.total_amount, project.total_amount, projectId]
  );

  await db.query(
    `INSERT INTO transactions (project_id, type, amount, status, payment_intent_id)
     VALUES ($1, $2, $3, $4, $5)`,
    [projectId, 'ESCROW_DEPOSIT', project.total_amount, 'HELD', paymentIntent.id]
  );

  res.json({ success: true });
});
```

☐ Frontend: Add Stripe Elements for payment

☐ Test with card: 4242 4242 4242 4242
☐ Verify escrow funded in Stripe dashboard

DAY 5-6: DEPOSIT RELEASE (6-8 hours)

☐ Create deposit_requests table:
```sql
CREATE TABLE deposit_requests (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id UUID REFERENCES projects(id),
  amount DECIMAL(10,2),
  breakdown JSONB,
  status VARCHAR(50),
  verification_due_date TIMESTAMP,
  approved_at TIMESTAMP,
  transfer_id VARCHAR(255),
  created_at TIMESTAMP DEFAULT NOW()
);
```

☐ Create deposit request endpoint + approval flow
☐ Frontend: Deposit request form with breakdown
☐ Release funds via Stripe Transfer API
☐ Test: Request $11.5k deposit, approve, verify transfer

TIME: 7-10 days total for Sprint 5

SPRINT 6: RECEIPT VERIFICATION SYSTEM
Duration: 5-7 days | Priority: HIGH - Competitive advantage

OBJECTIVE: Contractors upload receipts, AI verifies automatically

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TASKS:

☐ Create receipts table for storing receipt data
☐ Build file upload system (image/PDF)

☐ Integrate with cloud storage (S3 or similar)
☐ Implement OCR using Google Vision API OR OpenAI GPT-4V
☐ Extract: merchant, date, amount, line items
☐ Validate: date within 7 days, amount reasonable, items match project
☐ Calculate verification score (0-100)
☐ Auto-approve if score >70, flag if lower
☐ Frontend: Upload interface with drag-drop
☐ Display verification results to homeowner

SIMPLIFIED APPROACH (Recommended for MVP):
Use OpenAI GPT-4 Vision - single API call extracts everything

TIME: 5-7 days

SPRINT 7: CONTRACTOR VETTING (SIMPLIFIED)
Duration: 3-4 days | Priority: MEDIUM

OBJECTIVE: License + Insurance + Background check only

TASKS:

☐ Create contractor application form:
- License number
- Insurance certificate upload
- Basic info

☐ Integrate Checkr API for background checks ($30/check)
☐ License verification (state API if available, or manual)
☐ Insurance verification (call or API)
☐ Auto-approve or flag for manual review
☐ 24-48 hour turnaround
☐ NO tiers, NO complex scoring - just approved/rejected

TIME: 3-4 days

SPRINT 8: AI MILESTONE VERIFICATION
Duration: 5-6 days | Priority: HIGH

OBJECTIVE: AI analyzes milestone photos, scores completion

TASKS:

☐ Photo upload for each milestone
☐ AI analysis (GPT-4 Vision):
- Is work complete per description?
- Quality assessment
- Safety issues visible?
- Percentage complete estimate

☐ Generate AI score 0-100
☐ If score >85: Auto-approve (or send to lender)
☐ If score <85: Flag for manual review
☐ Frontend: Show AI analysis results
☐ Lender can override if needed

PROMPT EXAMPLE:
"Analyze these photos of a kitchen demolition milestone. The contractor claims demolition is 100% complete. Check for: all cabinets removed, walls exposed, debris removed, area ready for next phase. Provide completion percentage and quality score."

TIME: 5-6 days

SPRINT 9: FRONTEND COMPLETION
Duration: 5-7 days | Priority: CRITICAL

OBJECTIVE: Finish all 13 pages, polish UI

REMAINING PAGES (Based on 70% complete):

☐ Dashboard refinements
☐ Project detail page (100%)
☐ Milestone management (100%)
☐ Payment history page

☐ Receipt upload/verification page
☐ Contractor profile pages
☐ Settings pages
☐ Mobile responsiveness (all pages)
☐ Loading states
☐ Error handling
☐ Polish & design consistency

TIME: 5-7 days

SPRINT 10: TESTING & BUG FIXES
Duration: 3-5 days | Priority: CRITICAL

OBJECTIVE: Find and fix all critical bugs

TEST SCENARIOS:

☐ End-to-end homeowner flow:
- Create account
- Create project
- Fund escrow
- Approve deposit
- Approve milestones
- Review contractor

☐ End-to-end contractor flow:
- Apply to platform
- Get approved
- Connect bank
- Create project
- Request deposit
- Upload receipts
- Submit milestones
- Receive payments

☐ Edge cases:
- Insufficient escrow funds
- Rejected receipts
- Failed payments

- Disputed milestones

☐ Performance testing:
- Page load times
- API response times
- Large file uploads

☐ Browser testing:
- Chrome, Firefox, Safari
- Mobile (iOS, Android)

TIME: 3-5 days

SPRINT 11: PRE-LAUNCH PREP
Duration: 2-3 days | Priority: HIGH

OBJECTIVE: Everything ready for public launch

TASKS:

☐ Switch Stripe from test to live mode
☐ Set up production database
☐ Deploy to production server
☐ Configure domain & SSL
☐ Set up monitoring & error tracking
☐ Prepare support documentation
☐ Create onboarding videos
☐ Final security audit
☐ Backup systems tested
☐ Customer support email/chat setup

TIME: 2-3 days

SPRINT 12: LAUNCH!
Duration: 1 day | Priority: GO LIVE

OBJECTIVE: Platform live and accepting users

LAUNCH DAY CHECKLIST:

☐ Final smoke tests
☐ Announce to beta users
☐ Email Founder's Club members
☐ Activate lender pilot programs
☐ Post on social media
☐ Monitor for issues
☐ Respond to support requests
☐ Celebrate! 🎉

TIME: 1 day

TOTAL TIMELINE SUMMARY

Sprint 5: Stripe integration	7-10 days
Sprint 6: Receipt verification	5-7 days
Sprint 7: Contractor vetting	3-4 days
Sprint 8: AI milestone verification	5-6 days
Sprint 9: Frontend completion	5-7 days
Sprint 10: Testing	3-5 days
Sprint 11: Pre-launch	2-3 days
Sprint 12: Launch	1 day
TOTAL:	31-43 days

Best case: 4.5 weeks
Realistic: 6 weeks
Worst case: 7-8 weeks (with delays)

DAILY WORK SCHEDULE (Recommended)

Full-time focus (8 hours/day):
• Finish in 31 days (best case)

• Realistic: 40-45 days

Part-time (4 hours/day):
• Finish in 62 days
• Realistic: 80-90 days

Work schedule recommendation:
• Mornings: Deep development work
• Afternoons: Testing, documentation
• Fridays: Review week, plan next week
• Weekends: Optional catch-up time

PRIORITY RANKING (If Time-Constrained)

MUST HAVE (Cannot launch without):
1. Stripe escrow integration ⭐⭐⭐⭐⭐
2. Deposit release system ⭐⭐⭐⭐⭐
3. Milestone payments ⭐⭐⭐⭐⭐
4. Basic contractor vetting ⭐⭐⭐⭐⭐
5. Frontend completion ⭐⭐⭐⭐⭐

SHOULD HAVE (Launch faster without, add later):
6. Receipt verification ⭐⭐⭐⭐
7. AI milestone verification ⭐⭐⭐
8. Smart alerts ⭐⭐⭐

NICE TO HAVE (Post-launch additions):
9. Advanced analytics
10. Automated marketing
11. Mobile apps

QUICK REFERENCE: KEY COMMANDS

Start backend:
```bash
cd kealee-backend
npm run dev
```

Start frontend:

```bash
cd kealee-frontend
npm start
```

Database:

```bash
psql -d kealee_dev
```

Run migrations:

```bash
npm run migrate
```

Test Stripe webhooks locally:

```bash
stripe listen --forward-to localhost:3000/api/webhooks/stripe
```

GETTING HELP / RESOURCES

Stripe Documentation:
https://stripe.com/docs/connect

Stripe Connect guide:
https://stripe.com/docs/connect/enable-payment-acceptance-guide

OpenAI Vision API:
https://platform.openai.com/docs/guides/vision

Google Cloud Vision:
https://cloud.google.com/vision/docs

React documentation:
https://react.dev

PostgreSQL docs:
https://www.postgresql.org/docs/

LAUNCH READINESS CHECKLIST

TECHNICAL:

☐ All critical features working
☐ No critical bugs
☐ Performance acceptable (<3 sec page loads)
☐ Mobile responsive
☐ Production database configured
☐ Monitoring set up
☐ Backups automated

BUSINESS:

☐ Stripe live account approved
☐ Terms of service written
☐ Privacy policy published
☐ Pricing finalized
☐ Support system ready
☐ Marketing materials ready

TRACTION:

☐ 10 Founder's Club members signed up
☐ 2 lender pilot agreements
☐ 5 beta contractors ready
☐ 5 beta homeowners ready

WHEN ALL ☐ ARE ☑ → LAUNCH!

NEXT IMMEDIATE STEPS (THIS WEEK)

Day 1 (Today):

☐ Review this sprint plan
☐ Set up Stripe account
☐ Get test API keys
☐ Test Stripe connection

Day 2:

☐ Build contractor Stripe onboarding

☐ Test with 1 contractor account

Day 3:
☐ Build escrow funding
☐ Test with test credit card

Day 4:
☐ Build deposit request system
☐ Test deposit release

Day 5:
☐ Build milestone payment
☐ End-to-end test

By end of week: Escrow payments working! ✅

YOU CAN DO THIS TIM!

31-43 days until launch.
Focus on one sprint at a time.
Ship features, get feedback, iterate.

LET'S GO! 🚀